Article

SHIELD-MOEA: Disjoint Scenario Evaluation and Selective Screening for Proxy-Based Distribution-Network Resilience Planning

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Abstract

Distribution-network planning under load and outage uncertainty requires portfolios that remain useful when binding stresses change. SHIELD-MOEA combines constrained non-dominated sorting, feasibility repair, selective worst-scenario screening, and disjoint scenario draws for search and final evaluation. On a SimBench-derived benchmark with 72 candidate actions and five proxy objectives, its pooled mean hypervolume is 0.2740 over 30 seeds per stochastic method, 5.09% above NSGA-II with post-hoc repair and 5.56% above plain NSGA-II. Boundary runs retain positive within-setting margins under the tested budget, scenario-count, and survivability-coefficient settings, although clipping changes gap magnitude. Component controls identify feasibility repair as the resolved performance mechanism. Periodic re-screening and hybrid variation remain unresolved. Screening reduces static optimization-loop plan–scenario objective rows by 65%, excluding reference-bound construction and held-out evaluation, but neither front quality nor archived wall-clock time improves against no screening. An illustrative fixed-mapping AC check places SHIELD-MOEA within the feasible-rate range of the strongest repaired comparator rather than above it. The evidence supports a disjoint-evaluation workflow and a quality–workload diagnosis under the implemented load/outage proxies, not physical resilience superiority.

Keywords: distribution network planning; power system resilience; scenario screening; multi-objective evolutionary optimization; DER integration actions; NSGA-II; SimBench; AC power-flow mapping

1. Introduction

Resilience has moved from the margins of distribution planning to its center. Storm-driven outages, heat-wave load peaks, rooftop photovoltaics, and other distributed energy resources (DERs) create stress combinations not represented by historical load-duration curves. Reliability-report archives document the resulting operational and economic consequences [1,2]. At the same time, feeder reinforcement, storage, DER integration, and automation interact through a shared budget, and their value depends on the realized stress scenario. The planning decision is therefore inherently multi-objective: investment cost against losses, voltage exposure, reliability, and survivability under outages, all under scenario uncertainty.

Current methods leave a gap at the interface between the optimizer and the uncertainty model. Robust planning formulations can focus on a single worst case and scalarize the

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objective space, limiting visibility of the cost-resilience trade surface [3,4]. Evolutionary multi-objective methods expose that surface, but commonly evaluate candidates on a scenario set fixed before search [5,6]. The scenarios binding for mature plans, however, may differ from those affecting the initial population. When the same scenario sample steers and grades the search, sample-specific adaptation is also difficult to distinguish from robustness. Distribution utility planning departments instead require investment fronts that remain defensible under stress scenarios not used to tune the plans.

This paper proposes SHIELD-MOEA, a scenario-selective extension of non-dominated sorting for distribution resilience planning. The algorithm retains a standard multi-objective backbone–crowding-based selection, hybrid GA/DE variation, and greedy feasibility repair–and places its contribution at the optimizer–uncertainty interface. Every few generations, it identifies the K scenarios on which the current population performs worst and uses them for search-phase fitness. The final front is evaluated on a complete scenario set generated with disjoint random seeds, preventing direct reuse of search realizations in the reported score. We report mean and sampled worst-envelope hypervolume and supplement the proxy evaluation with an illustrative fixed-mapping pandapower AC check.

Real resilience-investment records are utility-internal, so evaluation uses a SimBench-derived proxy benchmark [7]. Statistics from the 18 highest-ranked subnets generate 72 candidate actions. The archive has eight experiment labels but only five active proxy configurations: four labels share the reference objective configuration under different optimizer seed streams, including `der_uncertainty`, because the configured DER-output multiplier is inactive in the p4 objective vector. The unseen-stress setting changes held-out load and outage ranges that do enter the proxy; its changed DER range does not. Shared infrastructure with CARS-MODE is declared in Section 2.4 and the Data Availability Statement.

The contributions of this paper are:

1. **A disjoint scenario-evaluation protocol.** SHIELD-MOEA uses separate draws for search guidance and final front scoring, preventing direct reuse of the realizations that shaped the population. Held-out and shifted-range diagnostics examine whether favorable proxy behavior persists beyond the search sample.
2. **Selective screening with an explicit workload–quality test.** Worst-K screening reduces static optimization-loop objective-row arithmetic from 51,200 to 17,920 under the declared schedule. The paired controls show no resolved front-quality or wall-clock improvement, distinguishing selective scenario exposure from a performance claim.
3. **Framework-level comparison with component-resolved evidence.** The complete pipeline is competitive against the implemented external controls across the archived and boundary settings, while feasibility repair accounts for the resolved internal gain. Hybrid variation, periodic re-screening, and survivability-only environmental selection remain unresolved, and the illustrative AC mapping bounds the transfer from proxy quality to electrical feasibility.

The evaluation consequently separates three questions: whether the complete pipeline is competitive, which component accounts for that result, and whether screening changes workload, solution quality, or both. This separation keeps disjoint evaluation, repair, and scenario screening from being treated as interchangeable sources of the observed proxy performance.

The paper proceeds as follows. Section 2 reviews related work. Section 3 defines the planning problem and the public benchmark. Section 4 describes SHIELD-MOEA. Section 5 gives the experimental protocol. Section 6 reports results, Section 7 discusses them, Section 8 states limitations, and Section 9 concludes.

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2. Related Work

SHIELD-MOEA draws on three bodies of literature that have so far evolved largely in parallel: resilience-oriented distribution planning under N-k and extreme-weather uncertainty, scenario generation and reduction for stochastic power system optimization, and hybrid multi-objective evolutionary algorithms with robustness-aware objectives. Below we discuss each thread and identify the gap that motivates this paper.

2.1. Resilience-Oriented Distribution Planning and Hardening Decisions

Panteli and Mancarella [8] distinguish high-impact, low-probability events from ordinary reliability and hardening from operational adaptation. Bhusal et al. [9] document the rapid growth of planning-stage resilience research. The dominant paradigm is robust optimization over an $N - k$ or weather-driven uncertainty set. Yuan et al. [3] formulate a two-stage robust problem with adversarial component outages. Ma et al. [5] extend this logic to joint hardening and distributed-generation placement with wind-dependent fragility, while Lin and Bie [6] add reconfiguration and DG islanding. Panteli et al. [10] connect weather intensity to component failure through fragility curves.

Practice-oriented studies consider resilient storage siting [11], zone-wise planning under decision-dependent uncertainty [12], and genetic-algorithm-driven feeder hardening [13]. Related work addresses resilient renewable microgrids [14] and islanding with fault recovery [15]. A parallel strand grounds planning claims in full AC physics. Mahmoudi and Alizadeh quantify errors caused by linearized expansion-planning models [16], while Trivic and Savic combine NSGA-II storage planning with power-flow analysis [17]. This evidence motivates the illustrative AC mapping check in Section 6.6.

A consistent pattern emerges across these studies. The trilevel robust models protect against a single worst case and collapse the objective space to one scalar, so the planner never sees the cost-resilience trade surface. The metaheuristic studies expose multiple objectives, but they evaluate candidate plans on a scenario set fixed before the search starts, and the same scenarios serve both to steer the search and to grade its output. None of them guards against a plan that has quietly overfitted the scenario sample it was optimized on.

2.2. Scenario Generation, Reduction, and Screening in Stochastic Power System Optimization

The second thread represents uncertainty through scenarios. Classical reduction selects a subset while minimizing probability distance; reference approaches include the Fortet–Mourier formulation [18] and forward/backward reduction [19]. Power-system applications adopted these methods early [20], and importance sampling made stochastic unit commitment tractable [21]. Generative models later reproduced spatiotemporal renewable behavior without parametric assumptions [22]. Applications include stochastic microgrid dispatch [23], distribution reconfiguration [24], and joint siting and sizing [25].

Recent work moves scenario machinery inside optimization. Li et al. learn an ambiguity set for distributionally robust DG planning [26], while Zhou et al. use extreme-event scenarios in integrated energy planning [27]. Decision-dependent uncertainty [28] is the closest mathematical-programming analogue to a scenario set that changes with candidate decisions. Wu et al. preserve extreme wind, PV, and load combinations during generation [29], paralleling the worst-K focus used here.

Most reductions are executed once, judged by distributional fidelity, and reused for every evaluation. The scenarios guiding search are therefore also used to grade the result. This distinction may be less important in a single monolithic solve. In population search, however, thousands of plans are compared while the binding scenarios can change with the population. SHIELD-MOEA separates search guidance from final scoring and tests whether updating the guidance subset adds value.

2.3. Hybrid Multi-Objective Evolutionary Algorithms and Robustness-Aware Objectives

The third thread concerns the optimizer. NSGA-II [30] and MOEA/D [31] remain common frameworks, with hypervolume [32] measuring front quality. Differential evolution [33] supplies mutation behavior complementary to GA crossover, and Das and Suganthan [34] review blended DE strategies. Jin and Branke [35] survey optimization under uncertain fitness, while Deb and Gupta [36] treat robustness as an objective over perturbation neighborhoods.

Recent work evaluates candidate solutions across fixed scenario sets [37] or combines decision preferences with variable-level uncertainty [38]. In both cases, the uncertainty set remains fixed during the run. SHIELD-MOEA tests periodic scenario-subset updates and, through a fixed worst-K control, whether those updates matter.

Recent power-planning studies hybridize evolutionary variation. Examples include an improved NSGA-II for storage-and-EV co-optimization [39], a three-stage TER-NSGA-II for resilient backbone-grid planning [40], and an enhanced beluga whale optimizer for storage planning [41]. Under uncertainty, such methods commonly average fitness over a fixed scenario sample or solve a robust reformulation before evolutionary search. The former incurs repeated evaluation cost, whereas the latter leaves the scenario set outside the evolving search state.

SHIELD-MOEA instead treats the evaluated scenario subset as a search-time quantity. The subset is periodically updated toward scenarios that are difficult for the current population. Evaluation on a disjoint scoring set then reports both mean and sampled worst-case hypervolume. The targeted controls in Section 6.3 test whether periodic updating adds value beyond a fixed generation-1 worst-K set.

2.4. Relation to CARS-MODE and Gap Statement

CARS-MODE [42] also studies evolutionary distribution planning on public benchmark-derived portfolios. It adapts DE control parameters and repairs budget violations in an economic DER-and-storage setting. SHIELD-MOEA instead studies the optimizer–uncertainty interface. Its NSGA-II-style kernel combines GA/DE variation and repair, while screening supplies search scenarios that remain disjoint from final scoring draws. The current p4 proxy responds to load and outage coordinates; its configured DER-output coordinate is inactive, although DER-oriented actions remain in the candidate pool and DER stress is present in the separate AC layer. The studies share the SimBench extraction, 72-action builder, binary-plan and budget representation, evaluation/statistics utilities, seeded archive machinery, and AC composition-mapping infrastructure. The companion's archived AC evidence covers four SimBench networks, whereas this paper's fixed-mapping extension covers those four networks plus CIGRE MV and IEEE 33-bus; neither panel is evidence for the other algorithm. Their research questions, method-specific search configurations, run archives, and conclusions remain separate.

The gap lies at the interface of search and uncertainty. Resilience studies often grade a robust solution or Pareto front on the same scenarios that guide optimization. Offline reduction is blind to the evolving population, while hybrid MOEAs usually adapt operators rather than exposure. SHIELD-MOEA combines worst-scenario screening, disjoint-draw final scoring, and sampled worst-case hypervolume reporting. The fixed worst-K control determines whether the adaptive update, rather than screening alone, is necessary.

3. Problem Formulation and Public Benchmark

3.1. Resilience-Oriented Planning as Scenario-Aware Multi-Objective Selection

Let $\mathcal{A} = \{a_1, \ldots, a_n\}$ be the candidate-action pool. Action a_i has synthetic cost c_i and coefficients $(l_i, u_i, h_i, r_i, g_i, d_i)$ for loss reduction, voltage-risk reduction, hosting-capacity gain, reliability gain, survivability gain, and a DER-readiness diagnostic. These coefficients are derived from public network statistics as described in Section 3.2. A plan is a binary vector $x \in \{0,1\}^n$. The configured scenario is $s = (\lambda, \delta, \sigma)$, where λ is a load multiplier, δ is a DER-output multiplier, and $\sigma \in [0,1]$ is outage severity.

For plan x under scenario s, the benchmark defines

$$
\begin{aligned}
L(x,s) &= \max\left(0.015, L_0\lambda - \frac{1}{120}\sum_{i=1}^{n} l_i x_i\right),\\
U(x,s) &= \max\left(0.005, U_0\lambda - \frac{1}{10}\sum_{i=1}^{n} u_i x_i\right),\\
H(x,s) &= \min\left(1, \frac{\sum_{i=1}^{n} h_i x_i}{\max(1, H_0\delta)}\right),\\
R(x) &= \min\left(1, 0.35 + \frac{1}{28}\sum_{i=1}^{n} r_i x_i\right),\\
S(x,s) &= \min\left(1, 0.42(1-\sigma) + \frac{1}{24}\sum_{i=1}^{n} g_i x_i\right).
\end{aligned}
$$

Here, for total selected-source load P_Σ and line length D_Σ, the implementation sets $L_0 = 0.12 + 0.015D_\Sigma / \max(1, P_\Sigma)$, $U_0 = 0.18 + 0.010P_\Sigma / \max(1, D_\Sigma)$, and $H_0 = 0.45P_\Sigma$. L is a loss index, U is a voltage-risk index, H is a hosting diagnostic, R is a scenario-invariant reliability proxy, and S is a survivability proxy driven only by outage severity and selected actions. The experiment labeled `restoration_aware_evaluation` replaces L by L^e during both search and evaluation:

$$
L^e(x,s) = L(x,s)\left[1 + 0.30\sigma\left(1 - S(x,s)\right)\right],
$$

which couples loss exposure to plan survivability. With investment cost $C(x) = \sum_{i=1}^{n} c_i x_i$ and scenario set $\mathcal{S}$, the implemented p4 five-objective problem is

$$
\begin{aligned}
\min_{x\in\{0,1\}^n} \quad F(x) = \Big(&C(x),\ \mathbb{E}_{s\in\mathcal{S}}[L(x,s)],\ \mathbb{E}_{s\in\mathcal{S}}[U(x,s)],\\
&-R(x),\ -\mathbb{E}_{s\in\mathcal{S}}[S(x,s)]\Big)\\
\text{subject to} \quad & C(x) \le B.
\end{aligned}
$$

The budget B is the only hard constraint, and violation is measured as relative budget overrun. The p4 objective vector omits H, so δ does not affect p4 search objectives, held-out objectives, hypervolume, or compromise choice. The coefficient h_i is nevertheless implemented in H and enters the repair and Weighted Sum heuristics specified below; it is not an advertised p4 objective. DER-output variation becomes active only in the separate AC operating scenarios. Accordingly, L, U, R, and S are transparent engineering proxies rather than power-flow solutions, and the `der_uncertainty` planning label is not evidence of DER-output robustness.

3.2. Candidate Pool Derived from SimBench

Candidate attributes derive deterministically from the SimBench complete mixed dataset [7]. Eligible subnets are ranked by $p_j + 0.2\ell_j$, where p_j is total active load and ℓ_j is

Table 1. The eight archive labels. Ranges are uniform sampling intervals for (load multiplier lambda, DER multiplier delta, outage severity sigma).

Experiment label	lambda range	delta range	sigma range	Implemented p4 effect
deterministic_vs_scenario	0.95-1.25	0.7-1.3	0.00-0.25	reference proxy configuration
der_uncertainty	0.95-1.25	0.4-1.7	0.00-0.25	inactive-factor replicate: δ is not consumed by p4 objectives
load_uncertainty	0.85-1.45	0.7-1.3	0.00-0.25	widened load band
outage_contingency	0.95-1.25	0.7-1.3	0.10-0.55	elevated outage severity
restoration_aware_evaluation	0.95-1.25	0.7-1.3	0.10-0.55	L^e replaces L in search and evaluation
scenario_screening_efficiency	0.95-1.25	0.7-1.3	0.00-0.25	reference proxy configuration under a separate optimizer seed stream
pareto_quality	0.95-1.25	0.7-1.3	0.00-0.25	reference proxy configuration under a separate optimizer seed stream
unseen_stress_generalization	0.95-1.25 (search)	0.7-1.3 (search)	0.00-0.25 (search)	evaluation uses $\lambda \in [1.3, 1.6]$, $\delta \in [1.4, 1.9]$, $\sigma \in [0.4, 0.7]$; only the λ and σ shifts affect p4 scores

total line length; the highest 18 each supply reinforcement, storage, DER-integration, and automation actions. Reactive load and the extracted average loading field are not used by the candidate builder. Let m_j be line count, n_j load count, e_j renewable capacity, $\tau_j = p_j / \max(0.2, \ell_j)$, and $q_j = \max(0, 0.55p_j - e_j)$. In the attribute order $(c_i, l_i, u_i, h_i, r_i, g_i, d_i)$, the four exact implemented constructions are

$$\begin{aligned}
\text{reinforcement: } & (60 + 4.5\ell_j + 7p_j, 0.012\ell_j + 0.020\tau_j, 0.020 + 0.006\tau_j, 0.06q_j, 0.018m_j, 0.016m_j, 0.20), \\
\text{storage: } & (50 + 16\sqrt{p_j + 1}, 0.025\sqrt{p_j + 1}, 0.018\sqrt{\tau_j + 1}, 0.12q_j + 0.08p_j, 0.055\sqrt{n_j + 1}, 0.070 \\
\text{DER integration: } & (45 + 10\sqrt{p_j + 1}, 0.018\sqrt{p_j + 1}, 0.012, 0.18q_j + 0.10p_j, 0.020\sqrt{n_j + 1}, 0.028\sqrt{n_j + \\
\text{automation: } & (38 + 1.8m_j, 0.006m_j, 0.010\sqrt{\tau_j + 1}, 0.025q_j, 0.085\sqrt{m_j + 1}, 0.115\sqrt{m_j + 1}, 0.35).
\end{aligned}$$

The resulting pool contains $n = 72$ candidates under nominal budget $B = 920$. Every attribute traces to the listed public-data fields through these rules, but the pool is a proxy rather than a costed engineering catalog and its cost unit is synthetic (Section 8).

3.3. Scenario Model and the Eight Archive Labels

Each label fixes ranges for the three configured scenario coordinates and draws $|\mathcal{S}| = 16$ scenarios uniformly. The search draw uses NumPy seed 20260713 and the evaluation draw seed 20260714; these two matrices are fixed and common to all methods and optimizer runs under a label. Table 1 lists the eight archived labels and distinguishes configured changes from changes that actually enter the p4 objective vector.

Thus the eight labels represent five active proxy configurations: the reference configuration is repeated under four labels with method- and label-specific optimizer seeds, and four additional labels activate load widening, outage widening, the L^e objective, or

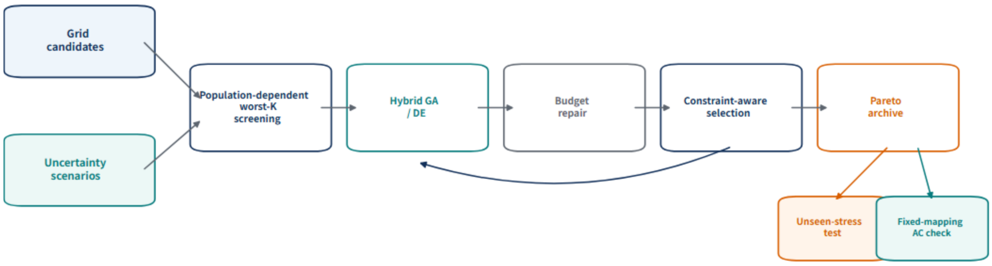

Figure 1. SHIELD-MOEA architecture. Scenario screening is population dependent; the feedback arrow denotes generational updating. The two evaluation branches use held-out stress scenarios or an illustrative AC mapping check and do not feed into selection.

held-out load/outage ranges. The last label tests distribution shift in the two active scenario coordinates; its non-overlapping δ range adds no p4 planning stress.

3.4. Shared Public Benchmark Statement

This paper and CARS-MODE [42] use a shared planning runner, SimBench subnet extraction and 72-action builder, binary-plan representation, budget handling utilities, fixed-bound hypervolume helpers, seeded archive/statistics machinery, and AC composition mapper. Four proxy coordinates are common; CARS-MODE uses hosting capacity as its fifth objective, whereas p4 uses survivability. The companion's archived AC panel contains four SimBench networks; p4 separately extends the fixed mapping to CIGRE MV and IEEE 33-bus for six networks in total. The paper-specific assets are the research question, method branch and controls, configurations, run archives, comparisons, AC evidence rows, and conclusions. Shared code and evaluation infrastructure do not make either paper's results evidence for the other.

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4. SHIELD-MOEA

Figure 1 closes the method's scenario-search loop. Each population induces a worst-K screened scenario subset, the hybrid variation operators generate candidate plans, budget repair restores feasibility, and robust environmental selection returns survivors. Unseen-stress and AC evaluations branch from the final archive and remain outside the optimization loop.

SHIELD-MOEA exposes screening, repair, outage information, and survivability in environmental selection as individual switches in the main ablation study. Additional within-SHIELD controls compare hybrid variation with GA-only and DE-only offspring and periodic re-screening with a fixed generation-1 worst-K subset. These controls support only the operator scopes stated in Sections 4.2–4.5. The method is a resilience-oriented extension of NSGA-II [30]: it retains constraint-dominated non-dominated sorting and crowding-distance selection, and adds hybrid variation, scenario screening, and feasibility repair.

Formal definitions. For binary plan x, cost feasibility is

$$C(x) = \sum_{j=1}^{n} c_j x_j, \qquad v_B(x) = \frac{\max(0, C(x) - B)}{B}.$$

Let l_q^{search} and u_q^{search} be fixed bounds generated on the search problem from the reference-plan sample in Section 5.3. At a screening generation, the implemented population-level scenario score is

$$\rho_t(s) = \frac{1}{|P_t|} \sum_{x \in P_t} \left[\sum_{q=1}^{5} \frac{f_q(x;s) - l_q^{\text{search}}}{u_q^{\text{search}} - l_q^{\text{search}}} + 10 v_B(x) \right].$$

The active subset is the worst K scenarios,

$$\mathcal{S}_t^K = \underset{\substack{\mathcal{A} \subseteq \mathcal{S}_{\text{search}} \\ |\mathcal{A}| = K}}{\arg\max} \sum_{s \in \mathcal{A}} \rho_t(s).$$

These screening normalizations are not clipped and use equal unit weights; cost and reliability add scenario-constant offsets, while λ and σ determine the scenario ordering and δ has no p4 effect. Between screening generations the active set is held fixed. Environmental selection uses the objective average over that set,

$$\bar{f}_q^K(x;t) = \frac{1}{K} \sum_{s \in \mathcal{S}_t^K} f_q(x;s).$$

The GA channel applies uniform crossover followed by independent bit mutation,

$$o_j^{GA} = \begin{cases} x_j^{(a)}, & U_j < 0.5, \\ x_j^{(b)}, & \text{otherwise}, \end{cases} \qquad o_j^{GA} \leftarrow 1 - o_j^{GA} \text{ with probability } 1/n.$$

For the DE channel, a relaxed donor is formed and mapped to a stochastic bit,

$$z = x_{r_1} + 0.5(x_{r_2} - x_{r_3}), \qquad o_j^{DE} \sim \text{Bernoulli}\big(0.08 + 0.82\, \mathbb{I}[z_j > 0.5]\big).$$

Repair uses the code's fixed, p4-independent candidate score

$$b_j = \frac{l_j}{0.12} + \frac{u_j}{0.02} + \frac{h_j}{5} + \frac{r_j}{2} + \frac{g_j}{2}, \qquad j^- = \underset{j: x_j = 1}{\arg\min} \frac{b_j}{c_j}, \qquad x_{j^-} \leftarrow 0,$$

repeating until $v_B(x) = 0$. Held-out evaluation forms both mean and componentwise-worst objective vectors,

$$f_q^{\text{mean}}(x) = \frac{1}{|\mathcal{S}_{\text{eval}}|} \sum_{s \in \mathcal{S}_{\text{eval}}} f_q(x;s), \qquad f_q^{\text{worst}}(x) = \max_{s \in \mathcal{S}_{\text{eval}}} f_q(x;s).$$

The mean-front plans are evaluated under both aggregations. Hypervolume alone applies clipped fixed-bound normalization, separately for mean and worst aggregations, followed by the standard dominated-volume definition,

$$HV_z(\mathcal{P}; r_z) = \lambda_Q \left(\bigcup_{x \in \mathcal{P}} \prod_{q=1}^{Q} [\tilde{f}_{q,z}(x), r_{q,z}] \right), \qquad z \in \{\text{mean}, \text{worst}\}.$$

Under the implemented schedule, static counting of objective-array rows implied by the code path gives

$$C_{\text{screened}} = G N_u K + N_r N_p |\mathcal{S}| = 12{,}800 + 5{,}120 = 17{,}920,$$

where union evaluation and ranking-round evaluation are disjoint terms. The implementation does not maintain a runtime call counter, so this is reproducible code-path arithmetic rather than an instrumented measurement or wall-clock claim.

4.1. Non-Dominated Sorting Kernel (NSGA-II Framework)

The kernel maintains a population of 40 binary plans over 40 generations. Environmental selection is constraint-dominated non-dominated sorting (feasible plans dominate infeasible ones; infeasible plans compare by relative budget violation) with crowding-distance truncation, following the standard NSGA-II procedure [30]. Initial plans are sparse random selections (3-18% density) and are repaired before generation 1 when repair is enabled. The baseline operator and repair-timing differences are stated in Section 5.1, so a framework-level margin is not attributed to the scenario interface alone.

4.2. Hybrid GA/DE Variation

Each generation produces 40 offspring. Half use uniform crossover followed by bit-flip mutation at rate $1/n$. The other half use the displayed DE-style relaxed donor followed by stochastic bit sampling with probabilities 0.9 and 0.08 above and below the strict 0.5 threshold. The three donor indices are sampled independently with replacement, so this channel is not a canonical distinct-index DE/rand/1 implementation. Section 6.3 compares this hybrid with controls that select GA-only or DE-only offspring while otherwise retaining the same SHIELD code path.

4.3. Worst-K Scenario Screening (The Adaptive Interface)

At generation 1 and every $T_s = 5$ generations thereafter, the algorithm evaluates the current population on each of 16 search scenarios, ranks scenarios by $\rho_t(s)$ (worst first), and retains the top $K = 4$. Until the next screening event, parent–offspring selection uses only the active set. Let $N_p = 40$, $N_u = 2N_p = 80$, $G = 40$, $|\mathcal{S}| = 16$, and $N_r = 8$ screening rounds (generations 1, 6, $\ldots$, 36). The unscreened path implies $C_{\text{full}} = GN_u|\mathcal{S}| = 51{,}200$ optimization-loop objective rows. The screened path implies disjoint terms $C_{\text{active}} = GN_uK = 12{,}800$ for union selection and $C_{\text{rank}} = N_rN_p|\mathcal{S}| = 5{,}120$ for ranking the current population, totaling 17,920. Thus the static optimization-loop count is 65% lower. This comparison excludes the reference-plan evaluations used to construct normalization bounds and all held-out final-front evaluation; those terms are outside the loop estimand and are not fully instrumented. Wall-clock behavior is evaluated separately.

4.4. Feasibility Repair

For SHIELD-MOEA, every initial and offspring plan exceeding the budget is repaired greedily with b_j/c_j before environmental selection. The NoRepair ablation skips both timings. The comparator labeled NSGA-II+Repair is different: it runs constrained NSGA-II without repair during search and applies the same drop rule only to the returned population. The ablation therefore identifies the implemented SHIELD repair schedule, while the NSGA-II comparison is rule-matched only at post-processing, not timing-matched.

4.5. Resilience Objective and Outage Awareness in Search

The survivability objective $-S(x, s)$ is part of environmental selection by default. `Ablation-NoResilienceObj` drops this column from parent–offspring environmental selection but, as implemented, the worst-K screening score ρ_t still uses all five objectives. Its result therefore tests removal from environmental selection, not complete removal of survivability information from search. `Ablation-NoOutage` instead sets $\sigma = 0$ throughout search, affecting both screening and environmental selection while leaving evaluation unchanged. The scope of each contrast is retained in Sections 6.3 and 6.6.

4.6. Direct-Reuse-Leakage-Controlled Evaluation and Worst-Envelope Rationale

No method–proposed, baseline, or ablation–is scored on the fixed scenarios it searched on. The evaluation scenario matrix is drawn with seed 20260714 rather than the search seed

20260713. In the unseen-stress label, the active load and outage ranges are also disjoint; the changed DER range is inert in the proxy. The final population is re-evaluated on this held-out matrix, filtered for budget feasibility, and reduced to a non-dominated front in raw objective space before any hypervolume normalization. Screening influences which plans are found but has no access to the held-out matrix.

We report two hypervolume variants. Standard mean hypervolume averages each objective over the 16 evaluation scenarios. The sampled worst-envelope readout reuses the plans on the mean-objective non-dominated front and replaces each scenario-dependent objective by its componentwise maximum over those 16 scenarios before hypervolume calculation; it does not construct or optimize a separate worst-objective front. This is a sample-dependent diagnostic rather than a bound on tail behavior. The two metrics use separate fixed normalization bounds (Section 5.3) and are comparable across methods only within their respective columns.

4.7. Algorithm Summary

```
SHIELD-MOEA(candidates, budget, search_scenarios, seed):
  initialize 40 sparse random plans; repair each to budget            # 4.4
  active <- search_scenarios
  for gen = 1 .. 40:
    if screening enabled and gen = 1 (mod 5):                         # 4.3
      score each search scenario by population-mean scalarized fitness
      active <- worst 4 scenarios
    offspring <- 20 by uniform crossover + bit-flip (1/n)             # 4.2
                 + 20 by the implemented DE-style stochastic donor
    repair offspring to budget                                        # 4.4
    F <- five-objective values of parents+offspring on active         # 4.5
    population <- constraint-dominated NDS + crowding (top 40)        # 4.1
  return final population
  wrapper re-evaluates it on the DISJOINT evaluation scenarios,
        filters budget-feasible plans, and extracts the raw-objective front # 4.
```

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5. Experimental Setup

5.1. Methods Compared

Table 2 lists eleven methods: SHIELD-MOEA, six baselines, and four single-switch ablations. Pymoo supplies NSGA-II and MOEA/D [43]. All methods see the same binary problem and fixed search-scenario matrix within a label, but their operators, constraint handling, returned object, repair timing, and–for MOEA/D–population size are not identical.

5.2. Protocol

Each stochastic method runs 30 independent optimizer seed indices $r \in \{0, \dots, 29\}$ per experiment label. The archive stores r, while the actual NumPy seed is reconstructed as

$$\mathrm{seed}(m, e, r) = 200000 + 7919r + (\mathrm{int}(\mathrm{SHA1}(\mathrm{p4}|e|m)_{1:6}, 16) \bmod 4096).$$

Because the method name enters the hash, methods do not share paired random-number streams; the Mann–Whitney tests are unpaired. Weighted Sum and Deterministic Planning ignore this seed: the uniform 2640-row archive retains 30 repeated invocations per label, but inference treats each rule as one effective result per label. The seed-level family contains eight stochastic opponents–four stochastic baselines and four ablations–and the

Table 2. Methods compared.

Method	Role	Description
SHIELD-MOEA	proposed	NSGA-II-style kernel + GA/DE hybrid variation + worst-K scenario screening + repair
NSGA-II	baseline	pymoo NSGA-II, population 40, sparse Boolean sampling, two-point crossover, bit-flip mutation, budget constraint, all 16 search scenarios
NSGA-II+Repair	baseline	the same constrained NSGA-II, followed by SHIELD's drop rule on the returned population only; no repair during search
MOEA/D	baseline	pymoo MOEA/D with 35 Das–Dennis reference directions, two-point crossover, bit-flip mutation, and $10^4 v_B$ objective penalty
GA	baseline	single returned point from tournament/uniform-crossover GA with mutation $1.5/n$, unclipped normalized-sum fitness plus $10 v_B$, and post-hoc repair
Weighted Sum	baseline	one deterministic greedy point ordered by b_j from Section 4.4 (not b_j/c_j and not a front-level weighted-sum solve)
Deterministic Planning	baseline	cost-first greedy fill under budget
Ablation-NoScenarioScreen	ablation	search on all 16 scenarios (screening off)
Ablation-NoRepair	ablation	feasibility repair disabled
Ablation-NoResilienceObj	ablation	survivability column removed from environmental selection; five-objective screening score retained
Ablation-NoOutage	ablation	outage severity zeroed during search; evaluation unchanged

sixteen deterministic gaps are descriptive. SHIELD-MOEA, NSGA-II, NSGA-II+Repair, and GA use population 40 and 40 generations. The configured MOEA/D uses 35 reference directions and hence 35 individuals for 40 generations, rather than the nominal population-40 value recorded at the top level of the JSON configuration.

5.3. Metrics and Statistics

The headline metric is standard hypervolume of the budget-feasible raw-objective non-dominated front on the evaluation scenario matrix, computed with pymoo's exact indicator at reference point 1.1 per normalized dimension. Objective bounds are computed once per active evaluation problem from a reference-plan sample generated with NumPy seed 20260713: the empty plan, all 72 single-action plans, and 2048 random plans greedily restricted to the budget, followed by a 5% span margin. No method output sets these bounds. Hypervolume normalization is clipped to $[0, 1]$; search scalarization and compromise selection use the corresponding affine values without clipping. The worst-envelope hypervolume evaluates the mean-front plans using per-objective maxima over the 16 evaluation scenarios and its own fixed bounds. An empty feasible front scores zero.

Statistical comparisons are two-sided Mann–Whitney U tests between SHIELD-MOEA and each of eight stochastic opponents per archive label ($n = 30$ per group), with Holm correction [44] within each label. Rank-biserial correlation and 5000-resample bootstrap intervals report effect size and mean-difference uncertainty. The deterministic rules receive point comparisons without seed-level p-values. All reported conclusions use the method-independent standard hypervolume defined below.

For reproducibility, the transformations between raw objectives, fronts, statistics, and the AC check are specified explicitly. The final front is first formed in raw objective space:

$$\mathcal{P}_{mer} = \{x \in \mathcal{X}_{mer} : C(x) \leq B,\ \nexists y \in \mathcal{X}_{mer} \text{ with } F(y) \prec F(x)\}.$$

For objective j, the evaluation reference sample supplies l_j and u_j. Hypervolume then uses

$$\widehat{f}_j(x) = \min\left\{1, \max\left[0, \frac{f_j(x) - l_j}{u_j - l_j}\right]\right\}.$$

With reference point $z^{\text{ref}} = (1.1, \ldots, 1.1)$, the primary score is the dominated union volume

$$HV(\mathcal{P}_{mer}) = \lambda_d\left(\bigcup_{x \in \mathcal{P}_{mer}} [\widehat{F}(x), z^{\text{ref}}]\right),$$

where λ_d denotes d-dimensional Lebesgue measure. The tail-oriented readout replaces each scenario-dependent objective by its observed maximum on the frozen evaluation scenarios,

$$f_j^{\text{wc}}(x) = \max_{s \in \mathcal{S}_{\text{eval}}} f_j(x, s),$$

before applying the same fixed-bound hypervolume construction to the same $\mathcal{P}_{mer}$ plans. The exported compromise is selected from the mean-objective front by an equal-weight affine sum without clipping,

$$x^{\star}_{mer} = \arg\min_{x \in \mathcal{P}_{mer}} \sum_{j=1}^{5} \frac{f_j(x) - l_j}{u_j - l_j}.$$

Only the counts of the four action kinds for $r = 0$ are exported to the AC mapper; nodal identities are discarded, and the first minimum is used if the array order contains a tie. For two seed samples of size n_1 and n_2, the reported rank statistic is

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - \sum_{i=1}^{n_1} R_i,$$

and the monotone Holm adjustment over the M within-label comparisons is

$$p_{(i)}^{\text{Holm}} = \max_{k \le i} \min\left\{1, (M - k + 1) p_{(k)}\right\}.$$

Finally, screening workload is reported independently of wall-clock time. Including both active-set union evaluation and full-set ranking rounds, the static optimization-loop code-path reduction relative to full-scenario union evaluation is

$$\eta_{\text{screen}} = 1 - \frac{G N_u K + N_r N_p |\mathcal{S}|}{G N_u |\mathcal{S}|} = 1 - \frac{17{,}920}{51{,}200} = 0.65.$$

This count and held-out front quality answer different estimands. The 65% reduction is analytic optimization-loop plan–scenario row arithmetic rather than an instrumented objective-call counter. Reference-bound construction and held-out evaluation are excluded from both loop counts; because final-front size varies and no call counter exists, the manuscript does not infer a full-pipeline reduction. Against no screening, the mean-HV contrast is unresolved in all eight labels, equivalence was not tested, and archived runtime is higher rather than lower. Every efficiency reading therefore retains this quality–workload asymmetry.

The separate AC layer declares a mapped case feasible only when all three engineering conditions hold:

$$I_{\text{AC}} = I(\text{converged})\, I(0.95 \le V_b \le 1.05\ \forall b)\, I(L_\ell \le 100\%\ \forall \ell).$$

5.4. Predeclared Boundary and Hypervolume-Audit Protocol

After reconciling the implemented equations, we froze a compact one-at-a-time matrix before execution. The reference setting uses budget factor 1.0, 16 search and 16 evaluation scenarios, $K = 4$, and the implemented survivability action term $\sum_i g_i x_i/24$. Two settings change the budget factor to 0.82 or 1.20; two change both disjoint draw sizes to 8 or 32 while preserving $K/|\mathcal{S}| = 1/4$; and two multiply only the selected-action term in the survivability equation by 0.75 or 1.25. The fixed repair score, including its $g_i/2$ term, is not rescaled, so the last axis is a formulation sensitivity rather than a repair sensitivity. The inactive DER-output multiplier remains generated but cannot affect any p4 objective.

For each of the seven settings, 30 independent method-specific seed streams run SHIELD-MOEA, NSGA-II+Repair, GA-only SHIELD, DE-only SHIELD, and fixed-worst-K SHIELD: $7 \times 5 \times 30 = 1050$ optimizer invocations. Search and evaluation use new disjoint scenario seeds 2026081301 and 2026081302. The NSGA-II comparator retains sparse binary sampling, two-point crossover, bit-flip mutation, the budget constraint, and final-population repair; the local executable uses the installed pymoo 0.4.1 implementation, so this boundary archive supplements rather than replaces the main comparison archive.

Each setting obtains one bounds vector before any method is run from the empty plan, all 72 singletons, and 2048 seeded budget-feasible random plans with the same 5% margin. A bounds digest is recorded on every run row, and verification requires one digest across all methods and seeds within the setting. Every raw mean-objective front is then scored three ways: the primary $[0,1]$-clipped hypervolume at $z^{\mathrm{ref}} = 1.1^5$, an unclipped audit at 1.1^5, and a predeclared alternative $[0,1]$-clipped score at 1.2^5. The primary contrast is SHIELD-MOEA minus NSGA-II+Repair; mechanism contrasts subtract each within-SHIELD control. Pointwise 95% percentile bootstrap intervals use 5000 fixed-seed resamples of the two independent method samples and are not multiplicity adjusted. Because each formulation setting has its own method-independent bounds, absolute HV levels are not compared across settings; only within-setting gaps are interpreted.

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6. Results

6.1. Proxy-Quality Comparison

The active-configuration structure precedes the pooled comparison. The archive contains five proxy configurations, while four labels are independent optimizer-seed blocks under the same reference ranges. Table 3 summarizes all 8 labels $\times$ 30 seed indices, and Figure 2 separates replicated reference blocks from configurations that change an active range or objective. The pooled view therefore gives the reference proxy configuration four times the weight of each other configuration; it is a descriptive archive summary rather than evidence from eight distinct uncertainty regimes.

SHIELD-MOEA attains pooled mean hypervolume 0.27396 (std 0.02700), 5.09% above NSGA-II with post-hoc final-population repair (0.26070) and 5.56% above plain NSGA-II. The corrected block-level contrasts favor the complete framework against the four stochastic baselines, but the repeated reference blocks do not create additional uncertainty regimes. Weighted Sum and Deterministic Planning are lower in the descriptive label-level comparisons and receive no seed-level tests. Component attribution is addressed separately in Section 6.3.

The display order makes the coverage boundary explicit. The first four rows use independent optimizer-seed streams but one active reference proxy configuration. The other four rows change the load range, outage range, loss objective, or held-out load/outage ranges. The `der_uncertainty` margin cannot be attributed to δ because that coordinate is

Table 3. Pooled proxy-quality comparison across eight archive-label blocks. Stochastic methods have 240 runs; deterministic rules summarize eight unique outputs retained as 240 repeated provenance rows. The eight blocks contain five active proxy configurations because the reference configuration appears four times.

Method	Role	Mean HV	Std	Worst-case HV	Mean front size	Mean runtime (s)
SHIELD-MOEA	**proposed**	**0.27396**	**0.02700**	**0.26911**	**38.9**	**0.089**
NSGA-II+Repair	baseline	0.26070	0.02696	0.25654	40.0	0.091
NSGA-II	baseline	0.25953	0.02816	0.25541	40.0	0.092
GA	baseline	0.21987	0.02811	0.22107	1.0	0.007
MOEA/D	baseline	0.00047	0.00002	0.00047	1.0	0.450
Weighted Sum	baseline	0.01963	0.00146	0.01805	1.0	0.0004
Deterministic Planning	baseline	0.02387	0.00124	0.02282	1.0	0.0004

Table 4. SHIELD-MOEA vs. the strongest external comparator, NSGA-II with post-hoc final-population repair. Intervals are pointwise, multiplicity-unadjusted 5000-resample bootstrap CIs for the mean HV difference; r_{rb} is rank-biserial correlation. Holm correction covers eight stochastic opponents per label and determines the significance statements.

Experiment	SHIELD	NSGA-II+Repair	ΔHV (95% CI)	r_{rb}	Holm p
deterministic_vs_scenario	0.26959	0.25322	0.01636 [0.01257, 0.02018]	0.893	1.46e-8
der_uncertainty	0.27137	0.25502	0.01635 [0.01334, 0.01955]	0.951	1.04e-9
scenario_screening_efficiency	0.26703	0.25495	0.01207 [0.00796, 0.01593]	0.744	3.80e-6
pareto_quality	0.26652	0.25349	0.01302 [0.00905, 0.01697]	0.738	4.77e-6
load_uncertainty	0.26211	0.24768	0.01443 [0.01046, 0.01841]	0.816	2.40e-7
outage_contingency	0.26982	0.25599	0.01382 [0.00995, 0.01747]	0.798	4.63e-7
restoration_aware_evaluation	0.33878	0.32699	0.01179 [0.00797, 0.01556]	0.718	7.44e-6
unseen_stress_generalization	0.24650	0.23824	0.00825 [0.00182, 0.01432]	0.442	0.0134

unused by p4 objectives. The unseen-stress label tests changed held-out load and outage ranges, not its simultaneous DER-output shift, and does not certify absence of scenario-distribution specialization.

Three baseline behaviors deserve honest annotation rather than silence. First, MOEA/D collapses on this problem (pooled HV 0.00047): with the budget handled as a penalty, its decomposition weights drive it to near-empty portfolios (mean selected actions 0.004). We report it as configured and note that a constraint-domination variant might fare better; the competitive comparison is carried by NSGA-II. Second, the single-solution methods (GA, Weighted Sum, Deterministic Planning) return one plan by construction, so their hypervolume measures the quality of a point, not a front. Third, runtimes are of the same order for the population methods (SHIELD-MOEA 0.089 s vs. NSGA-II 0.092 s per run at this instance size); no comparative runtime conclusion is drawn at this scale.

6.2. Sampled Worst-Envelope Readout

SHIELD-MOEA's pooled sampled worst-envelope HV is 0.26911, +5.36% above NSGA-II's 0.25541 and close to its mean-HV margin. Within the evaluation sample, no mean–envelope reversal is observed against this baseline. This diagnostic describes the 16 sampled scenarios; it is not a bound on unobserved tail behavior.

6.3. Repair Drives the Resolved Gain; Screening Reduces Optimization-Loop Rows

Figure 3 orders the single-switch evidence by what is resolved: repair first, followed by the screening quality–workload comparison, then the two narrower objective/exposure

Figure 2. Held-out proxy hypervolume (mean +/- std; 30 independent optimizer seeds per method and label) for SHIELD-MOEA, NSGA-II with post-hoc final-population repair, and plain NSGA-II. The first four blocks are identical-range replicates under one active proxy configuration; their independent optimizer seeds provide repeated samples, not four uncertainty regimes. The remaining blocks change an active range or objective. The DER-label block remains a reference replicate because δ is absent from p4 objectives.

switches. Pooled bars describe the eight archived blocks; the inferential decisions remain within-label.

The ablations separate the framework-level comparison from the contribution of individual mechanisms:

- **SHIELD's repair schedule is the main contributor.** Disabling it costs 8.10% of pooled hypervolume (0.25176 vs. 0.27396) and is Holm-significant in all 8 labels. Without repair, infeasible plans can occupy slots when too few feasible solutions are available, thinning the feasible population evaluated at the end.
- **Scenario screening has lower optimization-loop objective-row arithmetic without a resolved front-quality difference.** The screening-off ablation is nominally 0.48% lower and is unresolved in every label. The declared optimization loop evaluates 17,920 plan–scenario rows with screening and 51,200 without it, a 65% reduction after excluding reference-bound construction and held-out evaluation, but the implementation does not instrument an objective-call counter and equivalence was not tested. At this instance size, the full method is slower in the archived runtimes (0.0889 s versus 0.0792 s per run). The quality–workload asymmetry is therefore the result: fewer analytic loop rows, no detected quality gain, and no wall-clock saving.
- **Removing survivability from environmental selection does not yield a resolved quality difference.** The ablation yields pooled mean 0.27467, nominally +0.26% above the full method, and is not Holm-significant in any label (0/8; the label-level mean difference ranges from -0.0035 to +0.0039 in absolute HV). This is not a complete no-survivability-search ablation because the five-objective screening score remains active. The supported conclusion is limited to environmental selection: retaining the survivability column exposes a cost–survivability decision coordinate, but no search-quality gain is detected for that use.
- **Outage exposure has label-dependent proxy effects and an associational AC composition difference.** Under the primary within-label opponent families, zeroing outage severity is significant in `outage_contingency` ($p = 0.032$) and

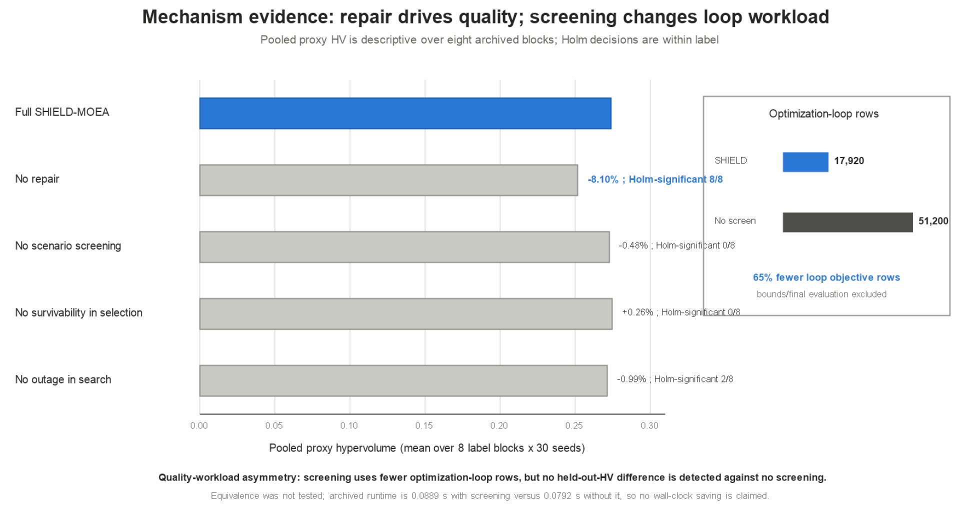

Figure 3. Left: pooled mean proxy hypervolume of SHIELD-MOEA and four single-switch ablations over 8 archive-label blocks $\times$ 30 seed indices, with relative differences and the number of within-label Holm-significant contrasts. "No survivability in selection" removes that column from environmental selection only; screening retains the five-objective scalar. Right: screening evaluates 17,920 static optimization-loop plan–scenario objective rows under the declared schedule versus 51,200 without screening. Reference-bound construction and held-out evaluation are excluded from both counts. This 65% loop-arithmetic reduction is paired with the unresolved quality contrast, the absence of an equivalence test, and the lack of a wall-clock saving.

`restoration_aware_evaluation` ($p = 0.009$). Under the supplementary outage-ablation family across eight labels, only `restoration_aware_evaluation` remains significant ($p = 0.024$; `outage_contingency` $p = 0.075$). The fixed-mapping AC layer is an illustrative composition comparison (Section 6.6), not a seeded causal effect.

Targeted variation and screening controls. Three additional controls use the same eight labels, 30 seed indices, population, generations, search/evaluation split, normalization, and evaluation schedule (720 runs). Pooled mean HV is 0.27425 for DE-only, 0.27408 for a worst-K set fixed after generation 1, 0.27396 for SHIELD-MOEA, and 0.27014 for GA-only. After Holm correction within each label, the full hybrid significantly exceeds GA-only in 3/8 labels and is inseparable in the other five; it is inseparable from DE-only and fixed worst-K in all eight. Thus the data support neither a hybrid-variation gain nor a dynamic-re-screening gain. The narrower interpretation is that the full stack is strong relative to the external baselines, repair is load-bearing, and screening has a lower static optimization-loop objective-row count; DE-only and one-shot worst-K are credible engineering simplifications on this benchmark.

Figure 4 shows the seed-level distributions for those controls. The pooled view is descriptive because label-specific scales differ; the formal verdict remains the within-label Holm analysis. It nevertheless makes both null results visible: the full method overlaps DE-only and fixed worst-K, while GA-only is lower in only part of the archive.

6.4. Parameter Sensitivity Analysis

We performed a one-at-a-time sensitivity analysis on the reference experiment (deterministic_vs_scenario). The sweep covers population size $N_p \in \{20, 40, 60\}$ (default 40, with NSGA-II rerun at each matched size), screening width $K \in \{2, 4, 8\}$ (default 4), and screening period $T_s \in \{3, 5, 10\}$ (default 5). Each point uses 10 independent seeds. Figure 5 and Table 5 summarize the results.

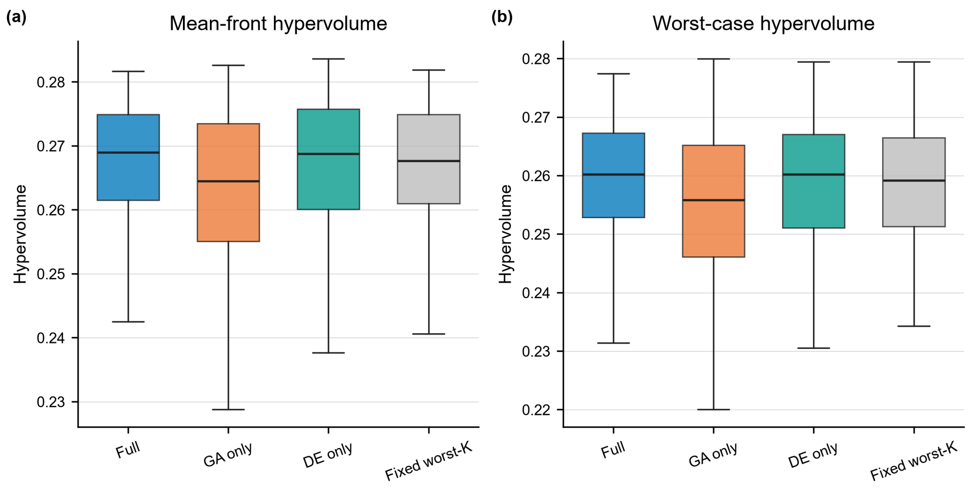

Figure 4. Seed-level mechanism controls across eight archive labels (240 runs per method). (a) Mean-front hypervolume; (b) sampled worst-envelope hypervolume. Boxes pool labels and seeds and are therefore descriptive. Within-label Holm tests show no separation between the full method and DE-only or fixed worst-*K* in 8/8 labels; the full method exceeds GA-only in 3/8.

Table 5. Exploratory parameter sweep (10 seeds per point). Reported Mann–Whitney p-values are nominal and multiplicity-unadjusted.

Parameter	Value	SHIELD-MOEA HV (mean +/- std)	NSGA-II reference	p (MWU)
population size N_p	20	0.2583 +/- 0.0155	0.2360	0.011
population size N_p	40 (default)	0.2674 +/- 0.0108	0.2527	0.007
population size N_p	60	0.2671 +/- 0.0120	0.2610	0.104
screening width K	2	0.2715 +/- 0.0062	0.2527	0.0003
screening width K	4 (default)	0.2658 +/- 0.0108	0.2527	0.014
screening width K	8	0.2701 +/- 0.0079	0.2527	0.001
screening period T_s	3	0.2688 +/- 0.0074	0.2527	0.0008
screening period T_s	5 (default)	0.2687 +/- 0.0085	0.2527	0.002
screening period T_s	10	0.2671 +/- 0.0121	0.2527	0.017

Note: the default configuration (population 40, K=4, T_s=5) appears once per sweep axis, so the table contains three default rows; they show slightly different SHIELD-MOEA means (0.2674, 0.2658, 0.2687) because each axis is drawn from an independent 10-seed stream – a normal consequence of sampling variance, not an inconsistency. For the same reason the NSGA-II reference is identical (0.2527) across the K and T_s rows, which reuse the matched population-40 reference, while the population-size rows re-run NSGA-II at each matched size (0.2360, 0.2527, 0.2610).

SHIELD-MOEA has a higher mean than NSGA-II at all nine parameter points, with a smallest absolute margin of 0.0061. Its mean varies by 5.0% of the default value across the sweep; the K and T_s axes vary by 2.1% and 0.6%, respectively. At $N_p = 60$, the raw comparison is unresolved ($p = 0.104$). Because the sweep has no confirmatory multiplicity family, the section supports only descriptive sensitivity patterns.

6.5. Predeclared Formulation Boundary and Hypervolume Audit

The immutable boundary archive contains all 1050 predeclared run rows, and its verifier finds one bounds digest per setting across every method and seed. Table 6 reports the primary within-setting comparison. Absolute HV values should not be read across rows because the formulation and its method-independent bounds change together.

SHIELD-MOEA has a positive primary mean gap over NSGA-II+Repair in every row, ranging from 0.01013 to 0.01885 (3.46–7.20% relative to the comparator), and all seven

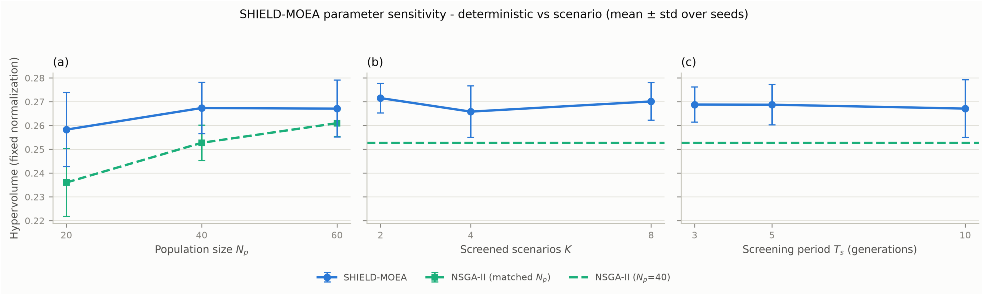

Figure 5. Parameter sensitivity over population size, screening width K, and screening period T_s (10 seeds per point; NSGA-II re-run at matched population sizes).

Table 6. Predeclared boundary reruns. Hypervolume uses fixed bounds within each setting, clipping to $[0,1]$, and reference point 1.1^5. Intervals are pointwise, multiplicity-unadjusted 5000-resample bootstrap intervals for the independent-sample mean difference. The final column counts all five methods and 30 seeds in that setting.

Setting	SHIELD	NSGA-II+Repair	ΔHV (95% CI)	Relative gap	Fronts clipped / 150
reference ($B = 1.00B_0$, $\lvert\mathcal{S}\rvert = 16, \gamma_g = 1.00$)	0.27108	0.25725	0.01383 [0.00963, 0.01809]	5.38%	13
tight budget ($0.82B_0$)	0.25909	0.24820	0.01090 [0.00765, 0.01441]	4.39%	149
loose budget ($1.20B_0$)	0.30269	0.29256	0.01013 [0.00557, 0.01457]	3.46%	77
8 scenarios ($K = 2$)	0.26986	0.25309	0.01677 [0.01288, 0.02047]	6.63%	20
32 scenarios ($K = 8$)	0.27255	0.25657	0.01598 [0.01082, 0.02083]	6.23%	17
survivability action scale $\gamma_g = 0.75$	0.28087	0.26201	0.01885 [0.01385, 0.02473]	7.20%	146
survivability action scale $\gamma_g = 1.25$	0.27263	0.25873	0.01390 [0.00993, 0.01793]	5.37%	19

pointwise intervals lie above zero. The sign persists when the scenario count changes from 8 to 32 and when the action term in S is scaled from 0.75 to 1.25. This is within-setting boundary evidence under new scenario and optimizer draws; it is not a pooled seven-regime effect and does not remove the main comparison's repair-timing caveat.

The hypervolume audit is consequential. Of 1050 fronts, 441 contain at least one normalized coordinate outside $[0,1]$; all 960 clipped coordinates are below zero, not above one. Clipping is nearly universal at the tight budget (149/150 fronts) and at $\gamma_g = 0.75$ (146/150). For those two settings, the SHIELD-minus-NSGA-II+Repair gap is 0.01090 versus 0.01559 without clipping and 0.01885 versus 0.02361 without clipping. Under the loose budget, clipping instead changes the gap from 0.00808 to 0.01013. Thus clipping does not reverse the comparison but is not magnitude-neutral.

At the alternative clipped reference point 1.2^5, the SHIELD-minus-NSGA-II+Repair gap remains positive in all seven settings; its relative range becomes 2.09–5.06%. Across all 28 full-method-minus-comparator contrasts (seven settings and four opponents), the mean-gap direction is identical under clipped 1.1^5, unclipped 1.1^5, and clipped 1.2^5. This sign audit is not an equivalence or invariance test: reference-point and clipping choices still change absolute and relative magnitudes.

Method independence also does not guarantee informative scaling. The reference-plan generator admits every singleton before checking feasibility. The resulting cost bounds are approximately -32,195 to 676,102 in every setting, whereas the tested budgets range only from 754.4 to 1104. This compresses all feasible cost differences into a narrow normalized band. The boundary results therefore support the direction of the within-setting comparator gap under the declared HV contract, but they do not establish that the current five-dimensional scale gives cost an engineering-calibrated influence.

Table 7. Illustrative fixed-mapping AC composition summary over six networks (108 cases per method). Only action-kind counts are transported; rows are descriptive mapped cases and carry no optimizer-level p-values.

Method	Role	AC-feasible rate	Stress-only feasible	Mean min Vm (pu)	Mean max loading (%)	Mean losses (MW)
NoPlan	reference	0.389	0.333	0.940	78.6	0.569
NSGA-II+Repair	baseline	0.694	0.633	0.969	52.7	0.373
Ablation-NoRepair	ablation	0.694	0.633	0.969	54.3	0.384
SHIELD-MOEA	proposed	0.685	0.622	0.969	52.7	0.377
Ablation-NoResilienceObj	ablation	0.685	0.622	0.970	56.0	0.384
NSGA-II	baseline	0.667	0.611	0.967	52.8	0.376
GA	baseline	0.667	0.600	0.969	48.1	0.362
Ablation-NoScenarioScreen	ablation	0.630	0.567	0.965	51.5	0.390
Ablation-NoOutage	ablation	0.574	0.533	0.969	67.0	0.463
Weighted Sum	baseline	0.417	0.333	0.945	73.7	0.537
Deterministic Planning	baseline	0.417	0.400	0.971	132.0	1.194
MOEA/D	baseline	0.389	0.333	0.940	78.6	0.569

Mechanism gaps remain unfavorable to a stronger attribution. DE-only has a higher primary mean than the hybrid in all seven settings, although every pointwise interval includes zero. Full SHIELD and fixed-worst-K are unresolved in all seven settings. GA-only is also unresolved except under the tight budget, where full-minus-GA-only is -0.00326 with a pointwise interval [-0.00582, -0.00061]. Because these 21 mechanism intervals are exploratory and multiplicity-unadjusted, the isolated adverse result is not a confirmatory superiority claim for GA-only; it is evidence against describing hybrid variation as uniformly beneficial.

6.6. Illustrative Fixed-Mapping AC Check on Six Distribution Networks

Hypervolume on proxy objectives cannot certify electrical feasibility. For three selected archive labels, the $r = 0$ equal-sum compromise is reduced to counts of reinforcement, storage, DER, and automation actions and mapped onto four SimBench MV networks (rural, semi-urban, urban, and commercial), the CIGRE MV benchmark, and the IEEE 33-bus feeder. Fixed rules parallel the most-loaded lines, attach storage to the weakest-voltage load buses at $\pm 3\%$ of net load, and add PV equal to 4% of net load at the highest-load buses. Automation has no steady-state effect. The original candidate identities are not transported, the identical mapping logic is used across network families, and no per-network optimizer tuning is performed.

Pandapower [45] solves AC power flow under base, 1.3x peak, 1.5x growth, 1.8x extreme growth, high-DER backfeed, and 1.5x growth with an $N-1$ outage. The selected planning labels are `deterministic_vs_scenario`, `der_uncertainty`, and `outage_contingency`; the middle label supplies a separate $r = 0$ optimizer composition but is not a DER-aware planning result because δ is inactive in p4. DER-output stress enters here through the high-DER AC operating case. A case is feasible when power flow converges, voltages remain within $[0.95, 1.05]$ pu, and line loading does not exceed 100%.

Each of the twelve planning/reference configurations faces $3 \times 6 \times 6 = 108$ cases, for 1296 AC solves in total. The readout is an illustrative, associational composition-level check under common placement rules, not a seeded replication of the proxy comparison, a nodal-plan validation, or a causal mechanism estimate. Table 7 reports the aggregate results; Section 6.8 separates the network families.

Three findings, in decreasing order of importance:

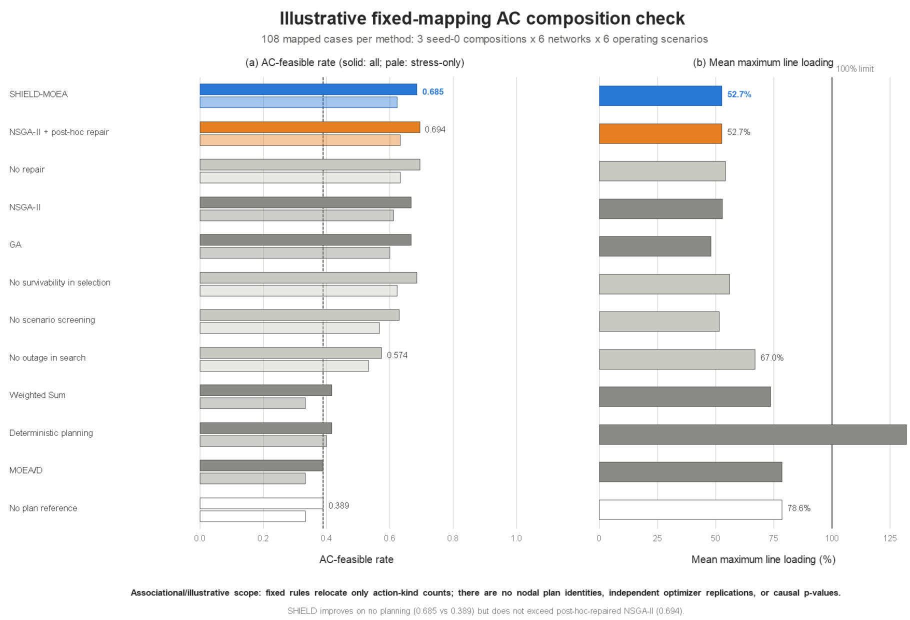

Figure 6. Illustrative fixed-mapping AC composition check over all six networks and 108 mapped cases per method. (a) AC-feasible rate over all cases and stress-only cases; the dashed line is the no-plan reference. (b) Mean maximum line loading; the vertical line is the 100% engineering limit. The figure transports seed-0 action-kind counts through fixed rules, so differences are associational and do not validate original nodal plans or identify causal operator effects.

1. **The mapped compositions improve aggregate feasibility but do not lead the field.** SHIELD-MOEA's compositions are AC-feasible in 68.5% of cases (62.2% under stress-only), against a no-plan reference of 38.9% (33.3% stress-only). Mean minimum voltage improves from 0.940 to 0.969 pu and mean maximum loading falls from 78.6% to 52.7%. Post-hoc-repaired NSGA-II is slightly higher in feasibility (69.4%), so no AC-superiority claim is made.
2. **The mapping associates outage-aware search with a different composition readout.** The NoOutage ablation–identical except that search never experiences outage severity–drops to an AC-feasible rate of 0.574 (0.533 stress-only), while mean maximum loading rises to 67.0%. This fixed-mapping composition difference is illustrative rather than a seeded causal estimate; it motivates a nodal rerun but does not establish an electrical mechanism effect.
3. **A cautionary baseline result.** Cost-first Deterministic Planning attains only a 0.417 AC-feasible rate and mean maximum loading 132.0%, because cheap actions add load-side assets without relieving binding corridors. Its loss value is also the largest in Table 7.

6.7. Screening Quality–Workload and Resilience-Composition Diagnostics

The pooled leaderboard alone does not answer whether scenario screening saves evaluation work or improves held-out quality. Figure 7 and Table 8 therefore place the two screening estimands side by side. The quality contrast uses archived held-out hypervolume; the workload contrast is deterministic code-path arithmetic under the declared schedule. They are intentionally not collapsed into a runtime-normalized score.

With $K = 4$ of 16 scenarios, active-set union evaluation is 75% smaller than full-scenario union evaluation. Eight full-set ranking rounds add 5120 objective rows, yielding an optimization-loop total of 17,920 rows versus 51,200 (65% lower). The common reference-bound construction and the variable-size held-out front evaluation are excluded, so this

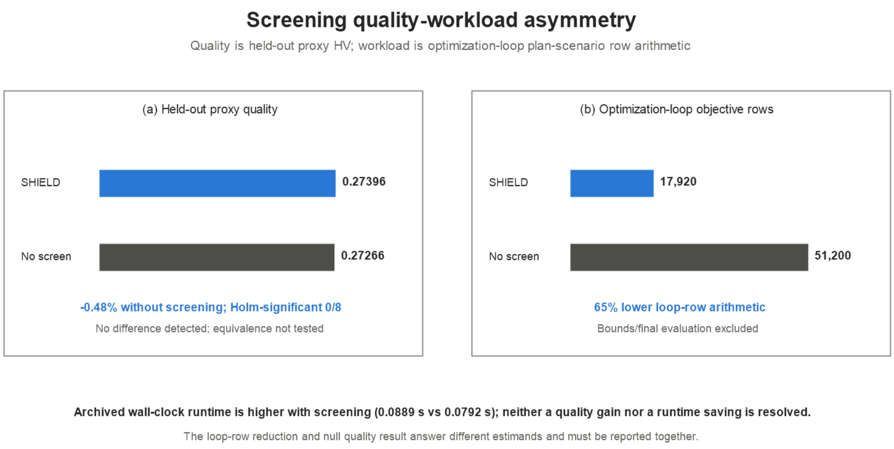

Figure 7. Screening quality–workload asymmetry. (a) Pooled held-out proxy hypervolume is 0.27396 for SHIELD-MOEA and 0.27266 without screening; the within-label contrast is unresolved in 8/8 labels and equivalence was not tested. (b) The declared optimization loop evaluates 17,920 static plan–scenario objective rows with screening and 51,200 without it. Reference-bound construction and held-out evaluation are excluded. The implementation lacks an instrumented objective-call counter, and archived wall-clock time is higher with screening, so neither a quality gain nor a runtime saving is claimed.

Table 8. Screening quality–workload record. Quality values are descriptive pooled means over eight archived blocks; inference is within label. Static objective rows are optimization-loop code-path counts, not an instrumented counter; reference-bound construction and held-out evaluation are outside this denominator.

Method	Mean proxy HV	Holm-significant labels vs. full	Optimization-loop objective rows	Mean runtime (s)	Supported reading
SHIELD-MOEA	0.27396	reference	17,920	0.0889	selective exposure with disjoint final scoring
NoScenarioScreen	0.27266	0/8	51,200	0.0792	no quality difference detected; equivalence not tested

is not a full-pipeline budget. Yet the quality contrast remains unresolved and archived runtime is 0.0889 s versus 0.0792 s. The efficiency result is therefore asymmetric: fewer analytic loop rows, no resolved quality gain, and no measured wall-clock saving. The row count cannot be promoted to energy or deployment savings without an instrumented expensive-simulator experiment.

Figure 8 separates the mean robustness score from its reliability and survivability components. SHIELD-MOEA has pooled reliability 0.9958, survivability 0.9940, and worst-case HV 0.2691. NSGA-II has slightly higher reliability (0.9972) but lower worst-case HV (0.2554); NoScenarioScreen has slightly higher survivability (0.9954) and a similar worst-case HV (0.2680). These crossed rankings are exactly why the paper does not describe the explicit resilience objective or dynamic screening as independently accuracy-improving mechanisms.

Taken together, the diagnostics define the supported mechanism claims. The full pipeline is competitive in proxy quality and has higher sampled worst-envelope HV than the tested stochastic baselines, but the current instance is too cheap for a wall-clock speed claim. Its supported elements are SHIELD's deterministic repair schedule, lower static scenario-row arithmetic, disjoint-draw scoring, and a descriptive outage-aware composition

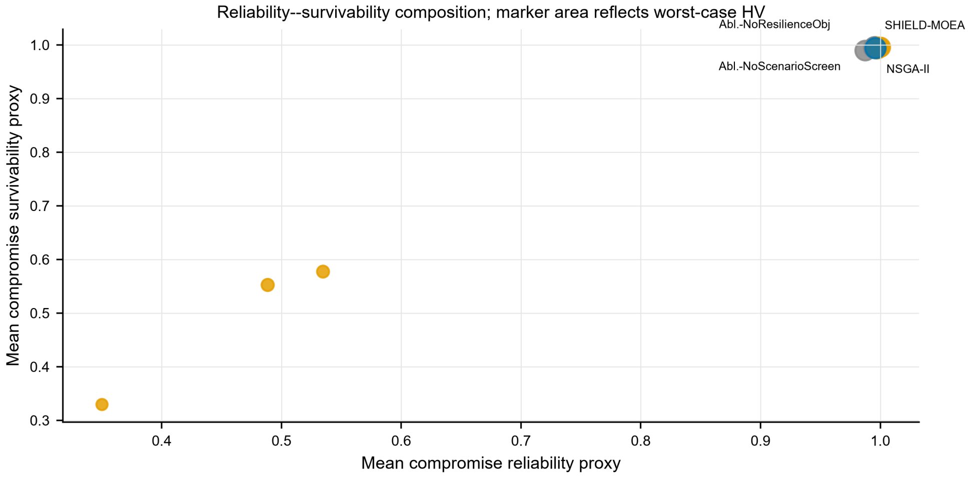

Figure 8. Pooled reliability and survivability of compromise portfolios derived from the frozen evaluation records; marker area represents worst-case hypervolume. The three readouts remain separately encoded rather than being collapsed into a synthetic index.

Table 9. Mechanism-level decision record.

Question	Evaluated comparison	Statistical result	Supported conclusion
Is SHIELD's repair schedule necessary?	SHIELD vs. NoRepair	significant in 8/8 labels	yes, for front quality in this stack
Is a front-quality difference detected for screening?	SHIELD vs. NoScenarioScreen	no significant difference in 8/8 labels	no difference detected; equivalence not tested
Is periodic re-screening better than a fixed worst-K set?	SHIELD vs. fixed generation-1 worst-K	no significant difference in 8/8 labels	not established
Does hybrid variation beat DE-only?	SHIELD vs. DE-only	no significant difference in 8/8 labels	not established
Do the mechanism conclusions change at the predeclared boundaries?	seven settings against GA-only, DE-only, and fixed worst-K	DE-only mean is higher in 7/7 but all pointwise CIs include zero; fixed worst-K is unresolved in 7/7; tight-budget GA-only has one adverse pointwise interval	no uniform hybrid or dynamic-screening benefit; no equivalence claim
Does outage exposure change the illustrative AC mapping readout?	SHIELD vs. NoOutage under fixed composition mapping	feasible rate 0.685 vs. 0.574	associational composition difference only; no seeded causal attribution

check. Hybridization and periodic re-screening remain unresolved design choices awaiting a harder simulator-backed benchmark.

6.8. Cross-Family AC Mapping Illustration

Figure 9 separates the illustrative fixed-mapping readout by network family. On the four SimBench networks, SHIELD-MOEA reaches 0.708 feasible rate. On CIGRE MV it reaches 0.944, compared with 0.167 for no planning, but post-hoc-repaired NSGA-II reaches 1.000. On IEEE 33-bus, both SHIELD-MOEA and post-hoc-repaired NSGA-II reach only 0.333, compared with 0.167 for no planning. The low absolute IEEE rate shows that a composition rule calibrated on SimBench does not automatically transfer to a radial feeder; it does not constitute independently optimized cross-family validation.

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7. Discussion

The complete pipeline and its components answer different questions. SHIELD-MOEA has higher pooled proxy hypervolume than the implemented external controls,

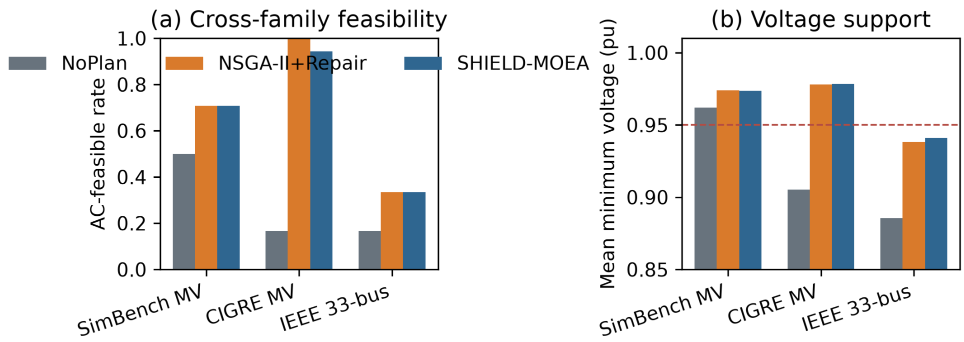

Figure 9. Illustrative AC-feasible rates and mean minimum bus voltage by network family for the no-plan reference, SHIELD-MOEA, and NSGA-II with post-hoc final-population repair. The dashed voltage line marks the 0.95 pu criterion. Each family receives the same three seed-0 action-kind compositions through fixed placement rules and six operating scenarios. SHIELD-MOEA improves over no planning in this mapping but does not outperform the comparator; no causal or nodal-transfer claim follows.

and the boundary archive retains positive within-setting margins under the tested budget, scenario-count, and survivability-coefficient settings. Four archived labels are replicated reference configurations rather than distinct uncertainty regimes, and clipping changes the size of the reported gaps. Within the SHIELD stack, feasibility repair accounts for the resolved gain. Hybrid GA/DE variation and periodic re-screening remain statistically inseparable from their simpler controls.

Scenario screening produces a workload–quality asymmetry. Screening reduces static optimization-loop plan–scenario rows from 51,200 to 17,920 under the declared schedule. Reference-bound construction and held-out evaluation are outside this denominator, no instrumented objective-call counter is available, and archived wall-clock time is not lower. The result therefore supports selective exposure to difficult scenarios without establishing a front-quality or runtime gain. Disjoint final evaluation remains valuable because it prevents the search sample from also serving as the grading sample.

Electrical mapping bounds the interpretation of the proxy front. The fixed-mapping AC compositions place SHIELD-MOEA within the feasible-rate range of the strongest repaired alternative. Removing outage exposure is associated with a lower feasible rate, but the comparison uses one reconstructed composition per method and is neither a seeded causal estimate nor a nodal-plan validation. The AC layer therefore identifies a transfer question rather than confirming physical resilience superiority.

For planning practice, the framework is a screening pipeline. It organizes constrained search, disjoint evaluation, and explicit workload accounting before nodal engineering analysis. Its most defensible contribution is not the necessity of every named operator, but the separation of framework performance, repair effects, screening workload, and electrical transfer. The companion CARS-MODE study [42] shares benchmark infrastructure but addresses a different operator question; the method-specific executions and conclusions remain separate.

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8. Limitations

Five limitation groups define the scope of the study.

1. **The archive labels do not represent eight independent resilience regimes.** The DER multiplier δ enters a hosting diagnostic omitted from the p4 objective vector, so neither the `der_uncertainty` label nor the δ shift in `unseen_stress_generalization`

changes proxy search, held-out hypervolume, or compromise choice. Four labels share the reference objective configuration; the pooled mean and 32/32 comparison count therefore weight that configuration four times. Electrical DER-output stress is evaluated only in the illustrative high-DER AC case.

2. **The proxy does not model restoration dynamics or nodal implementation.** The planning stage scales a survivability proxy but omits post-event restoration, reconfiguration, intentional islanding, and element-level fragility. The $N-1$ contingency appears only in the AC mapping layer. That layer transports action-kind counts from seed-index-0 compromises to six networks through fixed siting rules; it checks mapped compositions, not optimized nodal designs. Element-level $N-k$ sampling with restoration optimization and feeder-specific siting is required for a stronger resilience interpretation.
3. **Economic and benchmark external validity remains limited.** Candidate costs are uncalibrated analytic functions of network statistics, and all optimizer labels use one 72-candidate pool from one SimBench extraction. CIGRE MV and IEEE 33-bus expand only the illustrative AC layer; they are not independently optimized planning pools. Published utility-cost calibration and a second independently constructed candidate pool are therefore needed before economic or cross-benchmark claims.
4. **The component evidence is selective rather than universal.** The NSGA-II contrast is unresolved at population size 60 ($p = 0.104$); scenario screening is unresolved across all labels; hybrid variation is not separated from DE-only; and NoResilienceObj nominally exceeds the full method while retaining survivability in screening. Boundary intervals are pointwise and multiplicity-unadjusted. These results identify repair as the resolved component and leave screening, hybrid variation, and resilience-objective necessity open under the present protocol.
5. **Metric construction and comparator timing bound the headline gap.** Hypervolume uses engineering proxies and low-side clipping; 441/1050 boundary fronts require clipping, and the gap magnitude changes under the unclipped audit. The sampled worst envelope covers only 16 held-out scenarios and does not bound unseen tails. Moreover, the strongest NSGA-II comparator applies repair after optimization, whereas SHIELD repairs initialization and offspring before selection; the 5.09% result is a framework comparison, not an isolated timing effect. The supplementary boundary runner uses the disclosed pymoo 0.4.1 environment and is analyzed separately from the historical archive.

9. Conclusions

SHIELD-MOEA separates the scenario draws used to guide search from those used to score final fronts. On the SimBench-derived proxy, the complete pipeline has pooled mean hypervolume 5.09% above post-hoc-repaired NSGA-II and 5.56% above plain NSGA-II. Boundary settings preserve positive within-setting margins, although the archive contains replicated reference blocks and the reported gap magnitude depends on normalization and clipping.

The component evidence gives the framework result a narrower interpretation. Initialization-and-offspring repair accounts for the resolved internal gain. Screening reduces static optimization-loop plan–scenario rows from 51,200 to 17,920, excluding reference-bound construction and held-out evaluation, but no front-quality or wall-clock improvement is detected. Periodic re-screening does not outperform a fixed generation-1 worst-K set, and hybrid variation does not outperform DE-only. The illustrative AC mapping places

SHIELD-MOEA within the feasible-rate range of the strongest repaired comparator rather than establishing electrical superiority.

The contribution is therefore a disjoint-evaluation planning workflow, a resolved repair effect, and a measured workload–quality asymmetry for scenario screening. Post-event recovery, element-level contingencies, monetary calibration, independently optimized benchmark systems, and nodal AC evaluation remain necessary before stronger resilience-planning claims can be made.

Author Contributions

[AUTHOR INPUT REQUIRED: insert the final author list, assign CRediT roles, and obtain approval from every author.] All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

The source networks are public: candidate construction uses SimBench complete mixed network `1-complete_data-mixed-all-0-sw` (https://simbench.de), and the illustrative p4 AC mapping uses four SimBench MV networks, CIGRE MV, and IEEE 33-bus through pandapower (https://www.pandapower.org). This manuscript package contains the stage-4 canonical results-artifact manifest, its generator, four regenerated figures, four derived tables, and an output-hash build record. It also contains the stage-3 predeclared configuration, local executable, frozen 18-subnet source profile, 1050 boundary run rows, bounds and gap tables, and an immutable hash manifest. Stage 4 adds no experiment run. A complete executable supplement containing the shared main benchmark runner and imported helper, all historical controlling JSON configurations, 2640 main records, 720 targeted-control records, inference/effect tables, sensitivity records, 1296 AC-case records, and control-run orchestration has not yet been assembled in this package; a persistent release URL/DOI and source-data terms require author confirmation. The deterministic rules are treated as $n = 1$ per label for inference. The shared-infrastructure boundary with CARS-MODE [42] includes the planning runner, source extraction and candidate builder, binary-plan/budget structures, evaluation/statistics utilities, seeded archive machinery, and AC mapping. The companion's archived four-network panel and p4's six-network extension retain separate evidence rows; the research questions, method-specific configurations, run archives, comparisons, and conclusions are paper-specific.

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During the preparation of this manuscript, the authors used an AI-assisted coding environment for software development and a large language model for text editing and readability improvement. The authors reviewed and revised all assisted content and take full responsibility for the intellectual content and conclusions of the publication.

Conflicts of Interest

The authors declare no conflicts of interest.

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